Code and branch coverage command output

Source commit: 93ef68f21d0fba21945c7cb0f675dc7b9ba9f667

Tracked source changes: no

Generated from actual command output; test counts in separate commands may overlap.

Frontend test output includes component, service and integration suites.

==

COMMAND: npm run test:coverage

Started UTC: 2026-11-06T14:46:36.403Z

> se-jackfruit-security-foundation@0.1.0 test:coverage

> vitest run --coverage

RUN v5.0.3 D:/SEMESTERS/SEM5_2025/SE-JACKFRUIT

Coverage enabled with v8

stdout | backend/tests/integration/team-sprint2.test.ts > serves real Quiz retrieval and trusted submissions for 10 concurrent Student sessions

Real Quiz load: 10 sessions, 20 requests, p90 130ms (paired retrieval + submission).

stdout | backend/tests/integration/team-sprint2.test.ts > serves real Quiz retrieval and trusted submissions for 50 concurrent Student sessions

Real Quiz load: 50 sessions, 100 requests, p90 457ms (paired retrieval + submission).

✓ backend/tests/integration/team-sprint2.test.ts (9 tests) 11894ms

✓ connects real Question creation, Quiz discovery, trusted submission and persisted attempt without answer leakage 360ms

✓ enforces actual Quiz moderation, excludes discovery/submission and preserves existing attempt history 339ms

✓ makes moderated Questions unavailable through real Quiz reads and trusted scoring 359ms

✓ rejects malformed Quiz filters, forged scores and Admin submissions 315ms

✓ deduplicates racing retries of one submission, rejects changed answers and permits a fresh repeat attempt 359ms

✓ retains history and retry identity after reconnect and after moderation without exposing post-submission keys through history 455ms

✓ supports the submitted SAD download path with the same authenticated PDF bytes and removal denial 463ms

✓ serves real Quiz retrieval and trusted submissions for 10 concurrent Student sessions 1678ms

✓ serves real Quiz retrieval and trusted submissions for 50 concurrent Student sessions 6869ms

stdout | backend/tests/integration/sprint2.test.ts > keeps normal protected operations within three seconds for 10 concurrent authenticated sessions

NFR scoped smoke: 10 sessions, 20 authenticated requests; p90=56ms

stdout | backend/tests/integration/sprint2.test.ts > keeps normal protected operations within three seconds for 50 concurrent authenticated sessions

NFR scoped smoke: 50 sessions, 100 authenticated requests; p90=267ms

✓ backend/tests/integration/sprint2.test.ts (12 tests) 7655ms

✓ keeps normal protected operations within three seconds for 10 concurrent authenticated sessions 856ms

✓ keeps normal protected operations within three seconds for 50 concurrent authenticated sessions 3651ms

✓ backend/tests/integration/admin-provisioning.test.ts (16 tests) 3756ms

✓ runs the real CLI only against disposable MongoDB, uses its database, exits cleanly and prints no secrets 1152ms

✓ refuses the CLI before connecting when production or confirmation is invalid 478ms

✓ exits cleanly after a database-connected CLI refusal without promoting a Student or leaking input 610ms

✓ uses the common login and server-verified RBAC for real Question Bank creation and persistence 355ms

✓ backend/tests/integration/quizzes.test.ts (17 tests) 3182ms

✓ backend/tests/integration/local-demo.test.ts (2 tests) 3054ms

✓ serves protected APIs and SPA routes, preserving sessions, questions and removal across actual MongoDB restarts 2941ms

✓ backend/tests/integration/security.test.ts (17 tests) 2639ms

✓ IT-M4-01 / TC-Auth-01..03 MongoDB-backed account integration (14)

✓ uses the same error for wrong passwords and unknown or disabled accounts 360ms

✓ backend/tests/integration/questions.test.ts (8 tests) 2297ms

✓ backend/tests/integration/question-moderation.test.ts (4 tests) 2219ms

✓ moderates the actual Question model, excludes both read APIs and preserves persisted attempt references 366ms

✓ requires current admin role, confirmation and valid identifiers; removal is atomic 300ms

✓ discovers integrated modules and rejects malicious query/body fields 319ms

✓ persists question moderation and sessions after runtime recreation and database reconnect 408ms

✓ backend/tests/integration/runtime.test.ts (8 tests) 1810ms

✓ backend/tests/integration/attempt-api.test.ts (8 tests) 1840ms

✓ backend/tests/integration/bulk-moderation.test.ts (3 tests) 1458ms

✓ backend/tests/integration/quiz-attempt.test.ts (4 tests) 1191ms

✓ backend/tests/integration/quiz-contract-cross-module.test.ts (3 tests) 1274ms

✓ backend/tests/integration/attempt-repository.test.ts (5 tests) 618ms

✓ backend/tests/unit/quiz-model.test.ts (14 tests) 616ms

(node:31816) [MONGOOSE] Warning: Mongoose:
`Document.prototype.validateSync()` is deprecated and will be removed in Mongoose 10. Use `Document.prototype.validate()` instead.

(Use `node --trace-warnings ...` to show where the warning was created)

(node:31816) [MONGOOSE] Warning: Mongoose:
`Document.prototype.validateSync()` is deprecated and will be removed in Mongoose 10. Use `Document.prototype.validate()` instead.

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(node:31816) [MONGOOSE] Warning: Mongoose:
`Document.prototype.validateSync()` is deprecated and will be removed in Mongoose 10. Use `Document.prototype.validate()` instead.

(node:31816) [MONGOOSE] Warning: Mongoose:
`Document.prototype.validateSync()` is deprecated and will be removed in Mongoose 10. Use `Document.prototype.validate()` instead.

✓ backend/tests/unit/user-repository.test.ts (16 tests) 589ms

✓ backend/tests/integration/csmq45-history-isolation.test.ts (6 tests) 691ms

✓ backend/tests/unit/question-model.test.ts (13 tests) 644ms

✓ backend/tests/unit/quiz-attempt-model.test.ts (8 tests) 557ms

✓ backend/tests/unit/password.test.ts (17 tests) 379ms

✓ backend/tests/unit/release-guards.test.ts (3 tests) 224ms

✓ backend/tests/unit/middleware.test.ts (15 tests) 135ms

✓ backend/tests/integration/csmq45-submission-acceptance.test.ts (9 tests) 73ms

✓ backend/tests/unit/attempt-history-route.test.ts (3 tests) 50ms

✓ backend/tests/unit/config.test.ts (12 tests) 10ms

✓ backend/tests/unit/csmq-24-question-validation.test.ts (15 tests) 11ms

✓ backend/tests/unit/submission-service.test.ts (10 tests) 12ms

✓ backend/tests/unit/csmq-28-consolidated-suite.test.ts (5 tests) 9ms

✓ backend/tests/unit/material-upload-validation.test.ts (8 tests) 6ms

✓ backend/tests/unit/question-validation.test.ts (9 tests) 8ms

✓ backend/tests/unit/attempt-validation.test.ts (7 tests) 6ms

✓ backend/tests/unit/student-safe-serializer.test.ts (5 tests) 6ms

✓ backend/tests/unit/quiz-validation.test.ts (8 tests) 6ms

✓ backend/tests/unit/quiz-filters.test.ts (8 tests) 4ms

✓ backend/tests/unit/question-filters.test.ts (9 tests) 4ms

✓ backend/tests/unit/trusted-evaluation.test.ts (7 tests) 5ms

✓ backend/tests/unit/evaluate-quiz.test.ts (7 tests) 4ms

Test Files 37 passed (37)

Tests 330 passed (330)

Start at 20:16:37

Duration 67.28s (tests 79%, import 19%, worker 1%, transform 1%)

Isolate 37 workers spawned · ~101ms startup each (spawn + environment, per file)

at least ~3.63s faster with isolate: false — reuses workers across files instead of one per file

JUNIT report written to D:/SEMESTERS/SEM5_2025/SE-JACKFRUIT/reports/junit.xml

% Coverage report from v8

```
-------------------|---------|----------|---------|---------|-------------------
File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------|---------|----------|---------|---------|-------------------
All files          |   97.53 |    92.02 |   98.91 |   97.85 |
 src               |   97.97 |    98.63 |     100 |   97.46 |
  app.ts           |     100 |      100 |     100 |     100 |
```

```
 config.ts      |    100 |     100 |    100 |    100 |
 db.ts          |     60 |     100 |    100 |     60 | 12-13
 errors.ts      |    100 |   95.65 |    100 |    100 | 22
 runtime.ts     |    100 |     100 |    100 |    100 |
src/admin       |   87.5 |   83.33 |  85.71 |  85.18 |
 moderation.ts  |   87.5 |   83.33 |  85.71 |  85.18 | 27-30,41
src/attempts    |  95.91 |   86.74 |    100 |   96.5 |
 evaluate.ts    |    100 |     100 |    100 |    100 |
 history-route.ts |  91.66 |   83.33 |    100 |  91.66 | 46
 history.ts     |      0 |       0 |      0 |      0 |
 ...repository.ts |  93.47 |   75.75 |    100 |  95.23 | 46,124
 router.ts      |    100 |     100 |    100 |    100 |
 ...on-service.ts |  96.15 |   93.75 |    100 |  96.15 | 93
 ...evaluation.ts |     96 |   91.66 |    100 |     96 | 35
 validation.ts  |    100 |     100 |    100 |    100 |
src/materials   |  98.79 |      94 |  95.23 |    100 |
 upload.ts      |  98.79 |      94 |  95.23 |    100 | 31,88,98
src/middleware  |    100 |     100 |    100 |    100 |
 auth.ts        |    100 |     100 |    100 |    100 |
 validation.ts  |    100 |     100 |    100 |    100 |
src/models      |    100 |     100 |    100 |    100 |
 question.ts    |    100 |     100 |    100 |    100 |
 quiz-attempt.ts |   100 |     100 |    100 |    100 |
 quiz.ts        |    100 |     100 |    100 |    100 |
 user.ts        |    100 |     100 |    100 |    100 |
src/questions   |    100 |   91.66 |    100 |    100 |
 filters.ts     |    100 |     100 |    100 |    100 |
 router.ts      |    100 |      80 |    100 |    100 | 55-104
```

```
 serializer.ts   |    100 |     100 |    100 |    100 |
 validation.ts   |    100 |    87.5 |    100 |    100 | 71
src/quizzes      |  98.55 |   81.63 |    100 |  98.46 |
 filters.ts      |    100 |     100 |    100 |    100 |
 router.ts       |  97.95 |   71.42 |    100 |  97.77 | 23
 validation.ts   |    100 |   83.33 |    100 |    100 | 44
src/repositories |    100 |     100 |    100 |    100 |
 ...repository.ts |    100 |     100 |    100 |    100 |
src/routes       |    100 |     100 |    100 |    100 |
 auth.ts         |    100 |     100 |    100 |    100 |
src/security     |  96.42 |    92.2 |    100 |  98.48 |
 auth-service.ts |  93.33 |   92.59 |    100 |    100 | 14-22
 password.ts     |    100 |     100 |    100 |    100 |
 ...sion-admin.ts |  96.87 |   87.09 |    100 |     96 | 46
 sessions.ts     |    100 |     100 |    100 |    100 |
-----------------|--------|---------|--------|--------|-----------------

Finished UTC: 2026-11-06T14:47:44.964Z
Exit code: 0

================================================================================
========
COMMAND: npm run test:coverage --prefix frontend
Started UTC: 2026-11-06T14:47:44.964Z

> frontend@0.0.0 test:coverage
> vitest run --coverage
```

RUN v5.0.3 D:/SEMESTERS/SEM5_2025/SE-JACKFRUIT/frontend

Coverage enabled with v8

✓ src/pages/QuestionBankPage.test.tsx (9 tests) 7666ms

✓ QuestionBankPage (CSMQ-23 / CSMQ-24) (9)

✓ submits valid MCQ form and displays success message 2656ms

✓ ensures only results for current filters update the list when filter changes during save 2740ms

✓ clears loading state and displays results when an MCQ is saved while initial list is still loading 1756ms

✓ src/App.test.tsx (16 tests) 6689ms

✓ Study Commons routes and forms (16)

✓ keeps independent answers in the isolated Sprint 1 component 499ms

✓ shows a success message after successful registration 1176ms

✓ shows a duplicate-email error during registration 1193ms

✓ shows a useful error when registration cannot reach the server 1121ms

✓ shows login validation and API errors 906ms

✓ navigates to the student dashboard after successful login 853ms

✓ src/pages/MaterialsPage.test.tsx (15 tests) 3467ms

✓ submits PDF metadata through the shared authenticated service 785ms

✓ loads the 26th material and resets pagination when a new search is submitted 508ms

✓ preserves the submitted search after upload and resets the native PDF picker 332ms

✓ combines keyword and subject filters and explains unmatched results 438ms

✓ searches existing subjects in the filter menu and applies a selected subject 462ms

✓ src/pages/CreateQuizPage.test.tsx (10 tests) 3526ms

✓ CreateQuizPage (CSMQ-49) (10)

✓ validates empty subject on submission 428ms

✓ validates zero questions selected on submission 736ms

✓ submits valid quiz and displays success message 919ms

✓ displays API error message when creation fails 750ms

✓ tests/auth.integration.test.ts (3 tests) 1465ms

✓ connects the actual frontend client to registration, login, PDF storage and revoking logout 391ms

✓ tests/quiz-prototype.integration.test.tsx (2 tests) 1192ms

✓ IT-M3-02/03: renders Member 2 payload and evaluates mixed correct, incorrect and unanswered… from the actual UI 676ms

✓ IT-M3-02/03: renders Member 2 payload and evaluates all correct selections from the actual UI 515ms

✓ tests/admin-routing.test.tsx (4 tests) 1395ms

✓ routes successful Admin login to the dashboard with Question Bank and moderation links 1127ms

✓ src/pages/AdminPage.test.tsx (5 tests) 1141ms

✓ confirms that Delete all includes filtered-out items and cancellation sends no request 589ms

✓ offers per-item deletion with a separate confirmation and cancellation 303ms

✓ src/pages/QuizListPage.test.tsx (7 tests) 765ms

✓ src/pages/QuizAttemptPage.submit.test.tsx (6 tests) 393ms

✓ src/pages/AdminDashboardPage.test.tsx (4 tests) 211ms

✓ src/pages/QuizAttemptPage.session-expiry.test.tsx (2 tests) 240ms

✓ src/pages/QuizHistoryPage.test.tsx (4 tests) 266ms

✓ src/pages/QuizInsightsPage.test.tsx (4 tests) 213ms

✓ tests/admin-session.integration.test.tsx (3 tests) 252ms

✓ src/pages/QuizResultPage.test.tsx (6 tests) 173ms

✓ src/pages/QuizAttemptPage.result-navigation.test.tsx (2 tests) 189ms

✓ src/pages/QuizAttemptPage.csmq45-acceptance.test.tsx (2 tests) 208ms

✓ src/pages/StudentDashboardPage.test.tsx (2 tests) 160ms

✓ src/pages/QuizViewPage.test.tsx (3 tests) 168ms

✓ src/pages/QuizViewPage.navigation.test.tsx (2 tests) 164ms

✓ src/pages/QuizAttemptLoader.test.tsx (5 tests) 144ms

✓ src/pages/QuizHistoryPage.insights-navigation.test.tsx (1 test) 83ms

✓ src/services/authService.test.ts (19 tests) 54ms

✓ src/services/final-integration.test.ts (4 tests) 13ms

✓ src/services/quiz-security.test.ts (2 tests) 8ms

✓ src/services/quizAttemptDataService.test.ts (10 tests) 9ms

✓ src/services/attemptService.test.ts (8 tests) 7ms

✓ src/services/attemptHistoryService.test.ts (5 tests) 10ms

✓ src/services/attemptHistoryService.csmq45-boundaries.test.ts (3 tests) 6ms

✓ src/services/attemptInsights.test.ts (5 tests) 5ms

✓ src/services/quizDraft.test.ts (7 tests) 4ms

✓ src/services/recentMaterials.test.ts (2 tests) 8ms

✓ src/services/attemptInsights.csmq45-acceptance.test.ts (4 tests) 5ms

✓ src/services/attemptHistoryCsv.test.ts (4 tests) 5ms

Test Files 35 passed (35)

Tests 190 passed (190)

Start at 20:17:45

Duration 67.98s (tests 48%, environment 35%, setup 7%, import 6%, transform 2%, worker 1%)

Environment jsdom was created 35 times · 21.70s total, 35% of tracked time

create it once per worker with pool: 'vmThreads' (keeps per-file isolation) or isolate: false (shares it across files)

learn more: https://vitest.dev/guide/improving-performance#test-environments

JUNIT report written to D:/SEMESTERS/SEM5_2025/SE-JACKFRUIT/frontend/reports/junit.xml

% Coverage report from v8

```
-------------------|---------|----------|---------|---------|-------------------
File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------|---------|----------|---------|---------|-------------------
All files          |   78.45 |    71.86 |   79.27 |   81.46 |
 src               |   70.73 |    75.86 |    64.7 |   69.23 |
  App.tsx          |    72.5 |    75.86 |    64.7 |   71.05 | 72-85,127,143-161
  main.tsx         |       0 |      100 |     100 |       0 | 9
 src/components    |   55.55 |     58.2 |   56.52 |   60.46 |
  FormField.tsx    |     100 |      100 |     100 |     100 |
  ...alLibrary.tsx |       0 |        0 |       0 |       0 | 6-27
  Navbar.tsx       |   94.44 |    88.46 |     100 |    92.3 | 18
  ThemeToggle.tsx  |   88.23 |       80 |    87.5 |     100 | 11-19
 src/hooks         |   96.29 |    77.27 |     100 |     100 |
  ...ialLibrary.ts |   96.29 |    77.27 |     100 |     100 | 32-34,61-66
 src/pages         |   84.98 |    76.24 |   83.45 |   87.17 |
  ...boardPage.tsx |     100 |       80 |     100 |     100 | 18-20,32
  AdminPage.tsx    |   86.23 |    77.23 |   86.36 |   86.59 | ...32-133,152,171
  ...eQuizPage.tsx |   92.68 |     90.9 |   85.18 |    92.3 | 92-94,242-248,291
  HomePage.tsx     |     100 |      100 |     100 |     100 |
  LoginPage.tsx    |     100 |    83.33 |     100 |     100 | 11-13
  ...rialsPage.tsx |   75.11 |    66.12 |      70 |   82.67 | ...64,174,185,190
  ...nBankPage.tsx |   80.86 |    78.57 |   78.37 |   80.73 | ...38,363-364,409
  ...mptLoader.tsx |      88 |       75 |   88.88 |   95.45 | 112
  ...temptPage.tsx |   77.58 |    74.75 |   72.41 |   77.66 | ...19,486,600,642
  ...storyPage.tsx |   93.75 |    67.85 |   92.85 |   96.66 | 210
```

```
 ...ightsPage.tsx |  96.66 |   68.18 |  93.33 |    96 | 46
 QuizListPage.tsx |  89.13 |   78.72 |  91.66 |  89.13 | 74-80
 ...esultPage.tsx |  92.68 |   93.54 |    100 |  94.87 | 58,69
 QuizViewPage.tsx |    88 |   69.44 |    100 |    100 | 32-77,82-84,209
 RegisterPage.tsx |   100 |     90 |    100 |    100 | 11
 ...boardPage.tsx |   100 |   62.5 |    100 |    100 | 18-28
src/services      |  67.16 |   65.32 |  71.91 |   71.5 |
 ...HistoryCsv.ts |  47.61 |     50 |  83.33 |  41.17 | 33-44
 ...oryService.ts |  84.44 |   90.76 |   87.5 |    90 | 55,141,158,174
 ...ptInsights.ts |   100 |     75 |    100 |    100 | 30-31,71
 ...mptService.ts |    84 |   71.05 |    50 |   91.3 | 111-112
 authService.ts   |   75.6 |   64.4 |  67.74 |  78.94 | ...17-148,172-180
 ...Validation.ts |    80 |   84.37 |    100 |    100 | 7,9,11-16,18
 ...ionService.ts |  31.42 |   25.92 |    25 |  39.28 | 45-80,109-122
 ...ataService.ts |    80 |   75.71 |   87.5 |    80 | ...52,163-171,208
 quizDraft.ts     |   100 |    100 |    100 |    100 |
 quizService.ts   |   9.09 |   6.66 |   12.5 |  11.62 | 66-163
 ...tMaterials.ts |  77.77 |   82.35 |    100 |  91.66 | 19
-------------------|---------|----------|---------|---------|-------------------

Finished UTC: 2026-11-06T14:48:53.821Z
Exit code: 0

REPORT COMMAND EXIT CODE: 0
```